

1) (a) Derive the 2-point, open Newton-Cotes quadrature formula?

$$\int_a^b f(x) dx \approx I_{2, \text{open}}(f) = \frac{b-a}{3} \left( 2f(a+h) - f(a+2h) + f(a+3h) \right)$$

for evenly spaced points with  $h = \frac{b-a}{n+2}$

From notes:

$$\int_a^b f(x) dx \approx \sum_{i=0}^n w_i \cdot f(x_i)$$

For 2-points:  $h = \frac{b-a}{4}$ . Because we want the open formula, end-points will be excluded ( $x_0$  and  $x_4$ ). Nodes we'll use:

$$x_1 = a+h$$

$$x_2 = a+2h$$

$$x_3 = a+3h$$

We want the Lagrange Polynomials to replace our function

$$f(x) \approx P_n(x) = \sum_{j=0}^n f(x_j) \cdot L_j(x) \quad \int_a^b f(x) dx \approx \sum_{i=0}^n w_i \cdot f(x_i)$$

We'll use a trick for the Lagrange polynomials to simplify the integration by substituting  $x = a + t \cdot h \Rightarrow dx = \underline{h dt}$

The ends of the integral also change:

$$\text{for } x=a: a = a + t \cdot h \stackrel{h \neq 0}{\Rightarrow} \boxed{t=0}$$

$$\left. \begin{array}{l} \text{for } x=b: b = a + t \cdot h \\ h = \frac{b-a}{4} \Rightarrow 4h = b-a \Rightarrow b = a + 4h \end{array} \right\} \Rightarrow a + 4h = a + t \cdot h \Rightarrow \boxed{t=4}$$

So the nodes now will be ( $x = a + t \cdot h \Rightarrow t = \frac{x-a}{h}$ )

$$\bullet x_1 = a+h \rightarrow t_1 = \frac{(a+h)-a}{h} \Rightarrow \underline{t=1}$$

$$\bullet x_3 = a+3h \rightarrow \underline{t_3=3}$$

$$\bullet x_2 = a+2h \rightarrow t_2 = \frac{a+2h-a}{h} \Rightarrow \underline{t_2=2}$$

Our Lagrange Polynomials:  $L_{n,i}(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{(x-x_j)}{(x_i-x_j)}$

$$L_1(t) = \frac{(t-t_2)(t-t_3)}{(t_1-t_2)(t_1-t_3)} = \frac{(t-2)(t-3)}{+2} = \frac{1}{2}(t^2 - 5t + 6)$$

$$L_2(t) = \frac{(t-t_1)(t-t_3)}{(t_2-t_1)(t_2-t_3)} = \frac{(t-1)(t-3)}{-1} = -(t^2 - 4t + 3)$$

$$L_3(t) = \frac{(t-t_1)(t-t_2)}{(t_3-t_1)(t_3-t_2)} = \frac{(t-1)(t-2)}{2} = \frac{1}{2}(t^2 - 3t + 2)$$

Our weight functions:  $w_i = \int_a^b L_{n,i}(x) dx \rightarrow w_i = \int_0^4 h(t) L_{n,i}(t) dt$

$$w_1 = \int_0^4 h \cdot L_1(t) dt = \int_0^4 h \cdot \frac{1}{2}(t^2 - 5t + 6) dt = \frac{h}{2} \int_0^4 t^2 - 5t + 6 dt =$$

$$= \frac{h}{2} \left[ \frac{t^3}{3} - \frac{5t^2}{2} + 6t \right]_0^4 = \frac{h}{2} \cdot \left( \frac{64}{3} - \frac{5 \cdot 8}{2} + 24 \right) = \frac{h}{2} \cdot \left( \frac{64}{3} - 40 + 24 \right) = \frac{h}{2} \cdot \left( \frac{64 - 48}{3} \right) =$$

$$= \frac{h}{2} \cdot \left( \frac{16}{3} \right) = \underline{\underline{\frac{8h}{3}}}$$

$$w_2 = \int_0^4 h \cdot L_2(t) dt = \int_0^4 -h \cdot (t^2 - 4t + 3) dt = -h \cdot \left[ \frac{t^3}{3} - \frac{4t^2}{2} + 3t \right]_0^4 = -h \cdot \left[ \frac{t^3}{3} - 2t^2 + 3t \right]_0^4 =$$

$$= -h \cdot \left( \frac{64}{3} - 32 + 12 \right) = -h \cdot \left( \frac{64 - 60}{3} \right) = \underline{\underline{-\frac{4h}{3}}}$$

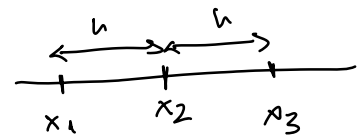
$$w_3 = \int_0^4 h \cdot L_3(t) dt = \int_0^4 \frac{h}{2} \cdot (t^2 - 3t + 2) dt = \frac{h}{2} \cdot \left[ \frac{t^3}{3} - 3 \frac{t^2}{2} + 2t \right]_0^4 = \frac{h}{2} \cdot \left( \frac{64}{3} - 24 + 8 \right) = \frac{h}{2} \cdot \left( \frac{64 - 48}{3} \right) = \underline{\underline{\frac{8h}{3}}}$$

Let's substitute these weights into the formula:  $\int_a^b f(x) dx \approx \sum_{i=0}^n w_i \cdot f(x_i)$

$$\begin{aligned} \int_a^b f(x) dx &\approx \frac{8h}{3} \cdot f(x_1) - \frac{4h}{3} \cdot f(x_2) + \frac{8h}{3} \cdot f(x_3) \\ &\approx \frac{4h}{3} \left( 2 \cdot f(x_1) - f(x_2) + 2 f(x_3) \right) \quad (?) \\ &\approx \frac{b-a}{3} \cdot (2f(a+h) - f(a+2h) + 2f(a+3h)) \end{aligned}$$

(b) Verify that this formula has degree of precision equal to 3.

Interval:  $[a, b] \rightarrow$  midpoint is  $x_2 = a+2h$   
use  $x_1 = x_2 - h$  &  $x_3 = x_2 + h$



Use Taylor's Expansion about the midpoint  $x_2$ :

$$f(x) = f(x_2) + f'(x_2)(x-x_2) + \frac{f''(x_2)}{2} (x-x_2)^2 + \frac{f'''(x_2)}{6} (x-x_2)^3 + \frac{f^{(4)}(x_2)}{24} (x-x_2)^4 + \dots$$

Our new interval will be  $[x_2-2h, x_2+2h]$

$$\int_{x_2-2h}^{x_2+2h} f(x) dx = \left[ f(x_2) \cdot x + \frac{f'(x_2)}{2} (x-x_2)^2 + \frac{f''(x_2)}{6} (x-x_2)^3 + \frac{f'''(x_2)}{24} (x-x_2)^4 \right]_{x_2-2h}^{x_2+2h} + \int_{x_2-2h}^{x_2+2h} \frac{f^{(4)}(\xi)}{24} (x-x_2)^4 dx$$

$$\begin{aligned} \cdot \frac{f''(x_2)}{6} (x-x_2)^3 &= \frac{f''(x_2)}{6} \cdot \left( (x_2+2h-x_2)^3 - (x_2-2h-x_2)^3 \right) \\ &= \frac{f''(x_2)}{6} \cdot ((2h)^3 - (-2h)^3) \\ &= \frac{f''(x_2)}{6} \cdot (8h^3 + 8h^3) \\ &= \frac{8h^3}{3} f''(x_2) \end{aligned}$$

$$\bullet f(x_2) \cdot x = f(x_2) \cdot (x_2 + 2h - (x_2 - 2h)) = 4h \cdot f(x_2)$$

For the error term, we apply the MVT theorem for Integrals:

$$\bullet \int_{x_2-2h}^{x_2+2h} \frac{f^{(4)}(\xi)}{24} (x-x_2)^4 dx \xrightarrow[u=dx]{u=x-x_2} \int_{-2h}^{2h} \frac{f^{(4)}(\eta)}{24} \cdot u^4 du = \frac{f^{(4)}(\eta)}{24} \cdot \left[ \frac{u^5}{5} \right]_{-2h}^{2h}$$

$$= \frac{f^{(4)}(\eta)}{24} \cdot \left( \frac{64h^5}{5} \right) = \frac{8h^5}{15} f^{(4)}(\eta)$$

$x=a: u=a-(a+2h) = -2h$   
 $x=b: u=(a+4h)-(a+2h) = 2h$

$$\int_a^b f(x) dx = 4hf(x_2) + \frac{8h^3}{3} f''(x_2) + \frac{8h^5}{15} f^{(4)}(\eta)$$

→ finite difference:

$$\begin{aligned} x_1 &= x_2 - h \\ x_3 &= x_2 + h \end{aligned} : f''(x_2) = \frac{f(x_1) - 2f(x_2) + f(x_3)}{h^2} - \frac{h^2}{12} f^{(4)}(\zeta) \quad (\text{from notes})$$

$$\int_a^b f(x) dx = 4hf(x_2) + \frac{8h^3}{3} \cdot \left( \frac{f(x_1) - 2f(x_2) + f(x_3)}{h^2} - \frac{h^2}{12} f^{(4)}(\zeta) \right) + \frac{8h^5}{15} f^{(4)}(\eta)$$

$$= 4hf(x_2) + \frac{8h^3}{3} \cdot \frac{f(x_1)}{h^2} - \frac{16h^3 f(x_2)}{h^2} + \frac{8h^3}{3} \cdot \frac{f(x_3)}{h^2} + \text{Error}$$

$$= \frac{8h}{3} f(x_1) - 4h f(x_2) + \frac{8h}{3} f(x_3) + \text{Error}$$

$$= \frac{4h}{3} \cdot [2f(x_1) - f(x_2) + f(x_3)] + \text{Error}$$

exact formula

**Reference:**

$$\int_a^b f(x) dx = 2hf(x_2) + \frac{h^3}{3} f''(x_2) + \frac{f^{(4)}(\eta)}{60} h^5$$

↳ finite difference approximation

$$f''(x_2) = \frac{f(x_1) - 2f(x_2) + f(x_3)}{h^2} - \frac{f^{(4)}(\zeta)}{12} h^2$$

⇒  $\int_a^b f(x) dx \approx \frac{h}{3} (f(x_1) - f(x_2) + f(x_3)) + O(h^5)$  ← add the two errors & apply MVT theorem again

for Error:  $h^5 \rightarrow O(h^5)$

So our degree of precision shows up for the 4th derivative, therefore

degree of precision is 3.

This is further verified from the fact that Newton-Cotes formula will give us  $n+1$  precision for  $n$  even and here  $n=2$ , then precision is 3. (\*) Last page

Remark: in my notes, I have written that I am supposed to apply the MVT when we add the errors, but I don't know how I would apply it here, so I just mimicked the steps from class.

2) Derive the composite midpoint with error:

$$\int_a^b f(x) dx = 2h \sum_{j=1}^n f(x_j) + \frac{(b-a)h^2}{6} \cdot f''(\xi)$$

where  $h = \frac{b-a}{2n}$  (step size),  $x_j = a + (2j-1) \cdot h$  &  $\xi \in [a, b]$   
midpoint?

Taylor Expansion about  $f(x_j)$ :

$$f(x) = f(x_j) + f'(x_j)(x-x_j) + \frac{f''(\xi)}{2} (x-x_j)^2$$

We'll integrate over  $[x_{j-h}, x_{j+h}]$  since  $x_j$  is our midpoint.

$$\int_{x_{j-h}}^{x_{j+h}} f(x) dx = \int_{x_{j-h}}^{x_{j+h}} x \cdot f(x_j) + f'(x_j)(x-x_j) + \frac{f''(\xi)}{2} \cdot (x-x_j)^2 dx$$

$$\bullet f(x_j) \cdot x = f(x_j) \cdot (x_{j+h} - (x_{j-h})) = f(x_j) \cdot 2h$$

MVT for Integrals on the error term:

$$\int_{x_{j-h}}^{x_{j+h}} \frac{f''(\xi)}{2} \cdot (x-x_j)^2 dx = \int_{-h}^h \frac{f''(\xi)}{2} \cdot u^2 du = \frac{f''(\xi)}{2} \left[ \frac{u^3}{3} \right]_{-h}^h = \frac{h^3}{3} \cdot f''(\xi')$$

So the integral over  $[x_{j-h}, x_{j+h}]$ :

$$\int_{x_{j-h}}^{x_{j+h}} f(x) dx = 2h f(x_j) + \frac{h^3}{3} \cdot f''(\xi')$$

So over the interval  $[a, b]$ :

$$\begin{aligned} \int_a^b f(x) dx &= \sum_{j=0}^n \int_{x_{j-h}}^{x_{j+h}} f(x) dx = \sum_{j=0}^n \left( 2h f(x_j) + \frac{h^3}{3} \cdot f''(\xi') \right) \\ &= \underbrace{2h \sum_{j=0}^n f(x_j)}_{\checkmark} + \sum_{j=0}^n \underbrace{\frac{h^3}{3} \cdot f''(\xi')}_{?} \end{aligned}$$

Based on notes online because I was genuinely stuck (link 3), we apply the Mean Value Theorem (the average of the derivatives is equal to the derivative at that point  $\xi' \in [a, b]$ )

$$\frac{1}{n} \cdot \sum_{j=0}^n f''(\xi') = f''(\xi_j)$$

$$\Rightarrow \sum_{j=0}^n f''(\xi') = n \cdot f''(\xi_j)$$

Back to the error term derived:

$$\frac{h^3}{3} \sum_{j=0}^n f''(\xi') = \frac{h^3}{3} \cdot n \cdot f''(\xi_j)$$

$$h = \frac{b-a}{2n} \Rightarrow n = \frac{b-a}{2h}$$

$$= \frac{h^3}{3} \cdot \left( \frac{b-a}{2h} \right) \cdot f''(\xi_j)$$

$$= \frac{h^2}{6} \cdot (b-a) f''(\xi_j)$$

So finally the composite midpoint formula with error:

$$\int_a^b f(x) dx = 2h \sum_{j=1}^n f(x_j) + \frac{(b-a)h^2}{6} \cdot f''(\xi_j)$$

3) Determine the weight  $w_i$  and quadrature points  $x_i$  for the three point Gauss-Legendre quadrature formula

$$\int_{-1}^1 f(x) dx \approx w_1 \cdot f(x_1) + w_2 \cdot f(x_2) + w_3 \cdot f(x_3)$$

What is the degree of this formula?

$$I_n(f) = \sum_{i=1}^n w_i \cdot f(x_i)$$

where

$$w_i(x) = \int_a^b L_i(x) \cdot \cancel{dx}$$

Three point Gauss-Legendre polynomial:  $P_3(x) = \frac{1}{2}(5x^3 - 3x)$

$$P_3(x) = 0 \Rightarrow 5x^3 - 3x = 0 \Rightarrow x(5x^2 - 3) = 0$$

$$x_2 = 0 \quad x_1 = -\sqrt{\frac{3}{5}} \quad x_3 = \sqrt{\frac{3}{5}}$$

Lagrange Polynomials for weight:

$$w_1(x) = \int_{-1}^1 \frac{(x-x_2)(x-x_3)}{(x_1-x_2)(x_1-x_3)} dx = \int_{-1}^1 \frac{x(x-\sqrt{\frac{3}{5}})}{(-\sqrt{\frac{3}{5}}-0) \cdot (-\frac{\sqrt{3}}{5}-\sqrt{\frac{3}{5}})} dx = \int_{-1}^1 \frac{x(x-\sqrt{\frac{3}{5}})}{\frac{6}{5}} dx =$$

$$= \frac{5}{6} \int_{-1}^1 x(x-\sqrt{\frac{3}{5}}) dx = \frac{5}{6} \left( \int_{-1}^1 x^2 dx - \int_{-1}^1 \sqrt{\frac{3}{5}} x dx \right) = \frac{5}{6} \cdot \left[ \frac{x^3}{3} \right]_{-1}^1 = \frac{5}{6 \cdot 3} \cdot (1^3 - (-1)^3) = \frac{5 \cdot 2}{18} = \frac{5}{9}$$

$$\begin{aligned} \cdot w_3 &= \int_{-1}^1 \frac{(x-x_1)(x-x_2)}{(x_3-x_1)(x_3-x_2)} dx = \int_{-1}^1 \frac{(x + \sqrt{\frac{3}{5}}) \cdot x}{(\sqrt{\frac{3}{5}} + \sqrt{\frac{3}{5}}) \cdot (\sqrt{\frac{3}{5}} - 0)} dx = \int_{-1}^1 \frac{x^2 + \sqrt{\frac{3}{5}}x}{\frac{6}{5}} dx = \\ &= \frac{5}{6} \int_{-1}^1 x^2 + \sqrt{\frac{3}{5}}x dx = \frac{5}{6} \cdot \left( \int_{-1}^1 x^2 dx + \int_{-1}^1 \sqrt{\frac{3}{5}}x dx \right) = \frac{5}{9} \end{aligned}$$

So :  $w_1 = \frac{5}{9}$  ,  $w_2 = \frac{8}{9}$  &  $w_3 = \frac{5}{9}$

$$x_1 = -\sqrt{\frac{3}{5}}, \quad x_2 = 0 \quad \& \quad x_3 = \sqrt{\frac{3}{5}}$$

The degree of precision is given by :  $2n-1$

for  $n=3 \rightarrow 2 \cdot 3 - 1 = \underline{\underline{5}}$



4) Consider the first-order initial value problem:

$$y' = y \cos t, \quad y(0) = 1 \quad t \in [0, 2\pi]$$

(a) Show there exists some interval  $I$  containing the initial point  $t=0$  on which a unique solution  $y(t)$  exists for  $t \in I$ .

Hint: it may be useful to prove that  $f(t, y) = y \cos t$  satisfies a Lipschitz condition).

From the well-posedness theorem if

i)  $f(t, y)$  is continuous in both  $t$  and  $y$  on some domain  $D \subseteq \mathbb{I} \times \mathbb{R}$

ii)  $f(t, y)$  satisfies a Lipschitz condition in the form:

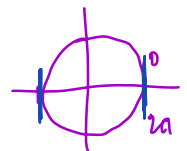
$$|f(t, y_1) - f(t, y_2)| \leq L |y_1 - y_2| \quad \text{for some constant } L > 0 \text{ \& } \begin{matrix} (t, y_1) \in D \\ (t, y_2) \in D \end{matrix}$$

Then there exists some interval  $J \subseteq \mathbb{I}$  containing  $t_0$ , such that the IVP has a unique solution on  $J$ .

So in our case  $f(t, y) = y \cos t$

i)  $f(t, y)$  is continuous since it is the product of two continuous functions (product rule for continuous functions)

ii) Lipschitz condition:  $J = \{ (t, y) \mid t \in [0, 2\pi], 0 \leq t \leq 2\pi \}$



$$|f(t, y_1) - f(t, y_2)| = |y_1 \cdot \cos t - y_2 \cdot \cos t| = |\cos t| \cdot |y_1 - y_2|$$

$$\cos(t) \text{ is bounded : } -1 \leq \cos t \leq 1 \longrightarrow 1 \cdot |y_1 - y_2|$$

for  $t \in [0, 2\pi]$

Therefore, since it satisfies both conditions, the IVP is well posed so it has a unique solution  $y(t)$  for  $t \in I$ .

(b) Carry out 2 steps of the forward Euler method (by hand) with step size of  $h = \frac{\pi}{4}$  to approximate  $y(\frac{\pi}{2})$ .  $y(0) = w_0 = 1$

$$\text{Formula for forward Euler: } w_{i+1} = w_i + h \cdot f(t_i, w_i)$$

$$\text{At } t_0 = 0: f(t_0, w_0) = w_0 \cdot \cos(t_0) = 1 \cdot \cos 0 = 1 \cdot 1 = 1$$

$$w_1 = w_0 + h \cdot f(t_0, w_0)$$

$$w_1 = 1 + \frac{\pi}{4} \cdot 1 \Rightarrow w_1 = 1 + \frac{\pi}{4}$$

$$\begin{aligned} \text{At } t_1 = t_0 + h = \frac{\pi}{4} : f(t_1, w_1) &= w_1 \cdot \cos(t_1) = \left(1 + \frac{\pi}{4}\right) \cdot \cos\left(\frac{\pi}{4}\right) \\ &= \left(1 + \frac{\pi}{4}\right) \cdot \frac{\sqrt{2}}{2} \end{aligned}$$

$$w_2 = w_1 + h \cdot f(t_1, w_1)$$

$$w_2 = \left(1 + \frac{\pi}{4}\right) + \frac{\pi}{4} \cdot \left(1 + \frac{\pi}{4}\right) \cdot \frac{\sqrt{2}}{2}$$

$$= \left(1 + \frac{\pi}{4}\right) + \frac{\sqrt{2} \cdot \pi}{8} \left(1 + \frac{\pi}{4}\right) \Rightarrow w_2 = \left(1 + \frac{\pi}{4}\right) \cdot \left(1 + \frac{\pi\sqrt{2}}{8}\right) \approx y\left(\frac{\pi}{2}\right)$$

(c) It can be shown that the exact solution  $y(t)$  is bounded by  $e^{-\pi} \leq y \leq e^{\pi}$  for  $(0 \leq t \leq 2\pi)$ .

Use this information to bound the maximum error of the approximate solution obtained by forwards Euler's method on  $t \in [0, 2\pi]$  as a function of step size  $h$ .

From the Global Discretization Error Bound for Explicit Euler Theorem, we can bound the maximum error with the

formula:  $|y_i - w_i| \leq \frac{h \cdot M}{2L} \cdot (e^{L(t_i - a)} - 1)$  where  $y_i = y(t_i)$

$L$  Lipschitz constant

$$\max_{t \in I} |y''(t)| \leq M, \text{ for some } M$$

$$y'(t) = y \cos t$$

$$\begin{aligned} y''(t) &= \frac{d}{dt} (y \cos t) = y' \cos t - y \sin t \\ &= (y \cos t) \cos t - y \sin t \\ &= y \cos^2 t - y \sin t \\ &= y (\cos^2 t - \sin t) \end{aligned}$$

We know that our exact solution is bounded  $e^{-\pi} \leq y \leq e^{\pi}$

$\Rightarrow \max |y''(t)| \leq M \Rightarrow \max |y (\cos^2 t - \sin t)| \leq e^{\pi}$ . We know that  $\cos^2 t$  is bounded by:  $0 \leq \cos^2 t \leq 1$  &  $\sin t$  is bounded by  $-1 \leq \sin t \leq 1$ , so the maximum of  $\cos^2 t - \sin t = \underline{2}$

$$\text{Thus, } \max |y (\cos^2 t - \sin t)| \leq 2e^{\pi} = M.$$

entire interval

Plug into formula:  $\text{Error} = |y_i - w_i| \leq \frac{h \cdot 2e^\pi}{2.1} (e^{1 \cdot (2\pi - 0)} - 1)$

$$\Rightarrow \text{Error} = |y_i - w_i| \leq h \cdot 2e^\pi (e^{2\pi} - 1)$$

(d) Carry out 2 steps of the Backwards Euler Method with step size of  $h = \frac{\pi}{4}$  to approximate  $y\left(\frac{\pi}{2}\right)$ .

Formula For Backwards Euler:  $w_{i+1} = w_i + h \cdot f(t_{i+1}, w_{i+1})$

$$\Rightarrow w_{i+1} = w_i + h \cdot (w_{i+1} \cdot \cos t_{i+1})$$

$$\Rightarrow w_{i+1} = w_i + h \cdot w_{i+1} \cdot \cos t_{i+1}$$

$$\Rightarrow w_{i+1} (1 - h \cdot \cos t_{i+1}) = w_i$$

$$\Rightarrow w_{i+1} = \frac{w_i}{1 - h \cdot \cos t_{i+1}}$$

For  $t_1$ :  $t_1 = 0 + h = \frac{\pi}{4}$  &  $w_0 = 1$  (from initial condition)

$$w_1 = \frac{1}{1 - \frac{\pi}{4} \cdot \cos\left(\frac{\pi}{4}\right)} = \frac{1}{1 - \frac{\pi\sqrt{2}}{8}}$$

For  $t_2$ :  $t_2 = t_1 + h = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$

$$w_2 = \frac{w_1}{1 - \frac{\pi}{4} \cdot \cos\left(\frac{\pi}{2}\right)} = \frac{w_1}{1 - 0} = w_1 = \frac{1}{1 - \frac{\pi\sqrt{2}}{8}}$$

(e) Find the exact solution to the IVP by separation of variables. Plot the exact solution, along with your approximate solutions from the Euler method. Comment on how the actual error compares to the theoretical bound for the forward Euler Algorithm.

$$y' = y \cos t \quad \text{I.C. } y(0) = 1$$
$$\frac{dy}{dt} = y \cos t \quad \uparrow$$
$$\frac{dy}{y} = \cos t \, dt \quad t=0$$

$$\int \frac{1}{y} dy = \int \cos t \, dt$$

$$\Rightarrow \ln|y| = \sin t + C$$

Using the initial condition:  $\ln(1) = \sin(0) + C$

$$0 = 0 + C \Rightarrow C = 0$$

$$\Rightarrow \ln|y| = \sin t \Rightarrow \boxed{y(t) = e^{\sin t}} \quad \text{exact solution}$$

$$\text{Actual Error: } E_{\alpha} = |y_{\text{exact}} - y_{\text{forward}}|$$

$$= \left| e^{\sin\left(\frac{\pi}{2}\right)} - w_2 \right|$$

$$= \left| e^1 - (\approx 2.7771) \right|$$

$$\approx \left| 2.7183 - 2.7771 \right| \approx 0.0588$$

And the theoretical Error from the (c) part:

$$E_{TH} \approx 1.834$$

The actual error is much smaller than the theoretical error bound. This happens because theoretical errors are the fail-safe errors, meaning its parameters are usually exacerbated in order to catch the worst-case scenario. The approximation with forward Euler is much more accurate and gives a more realistic bound.

5) Consider the IVP:  $y' = \sin(t) + y$ ,  $y(0) = 0$ ,  $t \in [0, 1]$

(a) Find the exact solution  $y(t)$ . Find the exact value of  $y(t)$  at  $t=1$

$$\frac{dy}{dt} = \sin(t) + y \Rightarrow \frac{dy}{dt} - y = \sin(t). \quad \text{Integrating factor: } \mu(x) = e^{\int 1 dt} = e^t$$

$$\Rightarrow e^{-t} \cdot \frac{dy}{dt} - e^{-t} \cdot y = e^{-t} \cdot \sin(t)$$

$$\Rightarrow \int \frac{d}{dt} (e^{-t} \cdot y) = \int e^{-t} \cdot \sin(t) dt$$

$$\Rightarrow e^{-t} \cdot y = \int \underbrace{e^{-t} \cdot \sin(t)}_I dt$$

$$I = \int e^{-t} \cdot \sin(t) dt \quad \text{Integration By parts: let } u = \sin(t) \Rightarrow du = \cos(t) dt$$

$$I = \overset{u \cdot v}{e^{-t} \cdot \sin(t)} - \overset{-\int v du}{\int (-e^{-t}) \cdot \cos(t) dt}$$

$$I = -e^{-t} \cdot \sin(t) + \int e^{-t} \cdot \cos(t) dt \quad \begin{array}{l} \text{Integration by Parts} \\ \text{again} \end{array} : \text{let } u = \cos(t) \Rightarrow du = -\sin(t) dt$$

$$dv = e^{-t} dt \Rightarrow v = -e^{-t}$$

$$\text{Substitute: } I = -e^{-t} \cdot \sin(t) + \left[ -e^{-t} \cdot \cos(t) - \int \underbrace{e^{-t} \cdot \sin(t)}_I dt \right]$$

$$\Rightarrow I = -e^{-t} \cdot \sin(t) - e^{-t} \cdot \cos(t) - I$$

$$\Rightarrow 2I = -e^{-t} (\sin(t) + \cos(t))$$

$$\Rightarrow I = -\frac{1}{2} \cdot e^{-t} (\sin(t) + \cos(t))$$

$$\text{So: } e^{-t} \cdot y(t) = -\frac{1}{2} \cdot e^{-t} (\sin(t) + \cos(t)) + C$$

$$\cdot e^t \Rightarrow y(t) = -\frac{1}{2} \cdot (\sin(t) + \cos(t)) + C \cdot e^t$$

$$\text{Apply init. cond: } y(0) = 0$$

$$0 = -\frac{1}{2} \cdot (\sin(0) + \cos(0)) + C \cdot e^0 \Rightarrow C = \frac{1}{2}$$

So the exact solution:

$$y(t) = -\frac{1}{2} \cdot (\sin(t) + \cos(t)) + \frac{1}{2} e^t$$

$$\Rightarrow y(t) = \frac{1}{2} (e^t - \sin(t) - \cos(t))$$

$$\text{For } t=1: y(1) = \frac{1}{2} \cdot (e - \sin(1) - \cos(1))$$

$$\Rightarrow y(1) = \frac{1}{2} \cdot (e - 1)$$

$$\Rightarrow y(1) \approx 0.6685$$

(c) The error plot implies linear convergence since it is a straight line on the loglog scale. Moving from  $h=10^{-1}$  to  $h=10^{-2}$  makes a drop of  $10^{-1}$  to  $10^{-2}$  on the error scale, so this repeated result shows us that it is of linear order  $O(h)$ .

(e) Both methods show linear convergence, meaning that the error is proportional to the step size. Backwards Euler needs a lot more computation time & resources since it is an implicit method. This means that at each step, our system needs to solve an equation to "guess" the next step, before actually determining the next value. Forwards Euler needs a lot less time & resources to compute since it is an explicit method.



If we are working with very large steps, forwards Euler may not be able to predict correctly, however I do think it also depends on the type of problem we're trying to solve.

In our case, I do think I could rearrange

⊛ We have our two error term say  $f'(a)$  &  $f'(b)$  on a bounded interval. Then there exists a max & min such that:

$$m \leq f'(a) \leq M$$

$$\Rightarrow 2m \leq f'(a) + f'(b) \leq 2M$$

$$\Rightarrow m \leq \frac{1}{2} (f'(a) + f'(b)) \leq M$$

$f'(c)$  from IVT.